Mini Project 2: Use the Android App designed by your seniors to captures the readings from motion sensors (accelerometer and gyroscope) in an android phone. Build a dataset using this application that contains the motion characteristics of an average person driving a two-wheeler. Your dataset must contain these variations: Multiple riders, multiple vehicles, different speeds and different phone placements. Build a model that generates a Driving Score/Risk Score indicating the driving behavior of an individual. Evaluate the trustworthiness for your model.

Bonus: Compare the performance of RNNs (LSTM/GRU etc.) vs. Transformers for the task.

Explanation of MODEL performance :-

1. Performance Comparison :-

LSTM

1. **Accuracy:** 0.84
2. **Best val_accuracy:** ~0.843
3. **Strengths:**
    a. Best overall balance
    b. Strong class-wise performance
4. **Weakness:**
    a. Slight confusion between adjacent classes (2–3, 3–4)

GRU

1. **Accuracy:** 0.84
2. **Best val_accuracy:** ~0.839
3. **Strengths:**
    a. Slightly faster than LSTM
    b. Better recall for some classes (e.g., class 4)
4. **Weakness:**
    a. Slightly less stable than LSTM

Transformer

1. **Accuracy:** 0.79
2. **Best val_accuracy:** ~0.80
3. **Strengths:**
    a. Very fast training ⚡
    b. Good at capturing global patterns
4. **Weakness:**
    a. Lower accuracy
    b. Struggles with time dependencies in your dataset

2. Observations :-

RNNs (LSTM/GRU)

- Designed for **sequential data**
- Your data = **time-series (sensor signals)** → perfect fit
- Capture:
  - Temporal dependencies
  - Driving patterns over time

Result: **Better accuracy (~84%)**

Transformers

- Designed for:
  - Large datasets
  - Parallel processing
  - Long-range dependencies

But in your case:

- Dataset may not be large enough
- Sequence length likely small
- No heavy contextual relationships

Result: **Lower accuracy (~79%)**

3. Training Speed Comparison :-

Model	Speed	Accuracy
LSTM	Medium	Best
GRU	Medium-Fast	Best-Moderate
Transformer	Fast	Moderate

4. Confusion Matrix Insight :-

Common issue across all models:

- Misclassification between:
  - Class 2 ↔ 3
  - Class 3 ↔ 4

Meaning:

- Driving behavior categories are close/similar
- Model struggles at boundaries (normal vs aggressive)

LSTM

```
Epoch 14/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 49s 20ms/step - accuracy: 0.8427 - loss: 0.3829 - val_accuracy: 0.8375 - val_loss: 0.3935
Epoch 15/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 49s 20ms/step - accuracy: 0.8456 - loss: 0.3755 - val_accuracy: 0.8408 - val_loss: 0.3929
Epoch 16/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 49s 21ms/step - accuracy: 0.8482 - loss: 0.3695 - val_accuracy: 0.8343 - val_loss: 0.4078
Epoch 17/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 82s 20ms/step - accuracy: 0.8506 - loss: 0.3608 - val_accuracy: 0.8408 - val_loss: 0.3897
Epoch 18/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 49s 20ms/step - accuracy: 0.8533 - loss: 0.3551 - val_accuracy: 0.8375 - val_loss: 0.3895
Epoch 19/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 49s 20ms/step - accuracy: 0.8554 - loss: 0.3482 - val_accuracy: 0.8398 - val_loss: 0.3952
Epoch 20/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 48s 20ms/step - accuracy: 0.8575 - loss: 0.3432 - val_accuracy: 0.8433 - val_loss: 0.3873
751/751 ━━━━━━━━━━━━━━━━━━━━ 5s 7ms/step
Classification Report:
              precision    recall  f1-score   support

           0       0.75      0.87      0.81      4567
           1       0.89      0.85      0.87      3764
           2       0.91      0.83      0.86      5631
           3       0.83      0.85      0.84      6496
           4       0.87      0.81      0.84      3558

    accuracy                           0.84     24016
   macro avg       0.85      0.84      0.85     24016
weighted avg       0.85      0.84      0.85     24016

Confusion Matrix:
[[3986  115   71  276  119]
 [ 226 3207  113  149   69]
 [ 359  111 4663  412   86]
 [ 424  118  267 5538  149]
 [ 297   40   37  286 2898]]
```

GRU

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Epoch 15/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 307s 119ms/step - accuracy: 0.8448 - loss: 0.3757 - val_accuracy: 0.8289 - val_loss: 0.4248
Epoch 16/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 85s 14ms/step - accuracy: 0.8470 - loss: 0.3688 - val_accuracy: 0.8383 - val_loss: 0.3946
Epoch 17/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 45s 19ms/step - accuracy: 0.8498 - loss: 0.3638 - val_accuracy: 0.8376 - val_loss: 0.3960
Epoch 18/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 48s 20ms/step - accuracy: 0.8518 - loss: 0.3593 - val_accuracy: 0.8390 - val_loss: 0.4027
Epoch 19/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 90s 23ms/step - accuracy: 0.8533 - loss: 0.3553 - val_accuracy: 0.8367 - val_loss: 0.3976
Epoch 20/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 33s 14ms/step - accuracy: 0.8535 - loss: 0.3504 - val_accuracy: 0.8394 - val_loss: 0.3959
751/751 ━━━━━━━━━━━━━━━━━━━━ 3s 4ms/step
Classification Report:
              precision    recall  f1-score   support

           0       0.76      0.87      0.81      4567
           1       0.88      0.86      0.87      3764
           2       0.90      0.81      0.85      5631
           3       0.89      0.82      0.85      6496
           4       0.78      0.87      0.82      3558

    accuracy                           0.84     24016
   macro avg       0.84      0.85      0.84     24016
weighted avg       0.85      0.84      0.84     24016

Confusion Matrix:
[[3981  112  119  111  244]
 [ 202 3253  124   75  110]
 [ 343  165 4568  352  203]
 [ 443  143  245 5330  335]
 [ 287    6   38  141 3086]]
```

Transformer

```
Epoch 14/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 9s 4ms/step - accuracy: 0.7987 - loss: 0.5027 - val_accuracy: 0.7941 - val_loss: 0.5092
Epoch 15/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 10s 4ms/step - accuracy: 0.8010 - loss: 0.4958 - val_accuracy: 0.7934 - val_loss: 0.5121
Epoch 16/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 10s 4ms/step - accuracy: 0.8030 - loss: 0.4908 - val_accuracy: 0.8026 - val_loss: 0.4915
Epoch 17/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 10s 4ms/step - accuracy: 0.8055 - loss: 0.4865 - val_accuracy: 0.7977 - val_loss: 0.5074
Epoch 18/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 10s 4ms/step - accuracy: 0.8076 - loss: 0.4824 - val_accuracy: 0.8026 - val_loss: 0.4969
Epoch 19/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 10s 4ms/step - accuracy: 0.8058 - loss: 0.4795 - val_accuracy: 0.7979 - val_loss: 0.4994
Epoch 20/20
2402/2402 ━━━━━━━━━━━━━━━━━━━━ 10s 4ms/step - accuracy: 0.8096 - loss: 0.4745 - val_accuracy: 0.7970 - val_loss: 0.5091
751/751 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step
Classification Report:
              precision    recall  f1-score   support

           0       0.77      0.78      0.77      4567
           1       0.86      0.78      0.82      3764
           2       0.84      0.75      0.79      5631
           3       0.71      0.87      0.78      6496
           4       0.90      0.77      0.83      3558

    accuracy                           0.79     24016
   macro avg       0.82      0.79      0.80     24016
weighted avg       0.80      0.79      0.80     24016

Confusion Matrix:
[[3541  143  210  605   68]
 [ 212 2950  183  383   36]
 [ 259  187 4228  901   56]
 [ 312  118  277 5628  161]
 [ 250   30  133  403 2742]]
```

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WARNING:absl:Compiled the loaded model, but the c
32/32 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step
32/32 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step
32/32 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step
Final Driving Score: 5
Behavior: Excellent
```

5. Final Verdict :-

1. Best Model: LSTM

- Highest validation accuracy
- Most stable learning
- Best suited for sensor time-series

2. Second: GRU

- Nearly same performance

- Faster → good for deployment

3. Third: Transformer

- Not ideal for this dataset (yet)

Final Conclusion

RNNs (LSTM/GRU) > Transformers

results (~84% accuracy) are very solid